

Name: _____

Roll No: _____

Program: BS (CS, AI)

Semester: Fall 2023

Time Allowed: 3:00 hour

Course: Discrete Structures

Examination: Final

Weight 55% Total Marks: 62

Date: Dec, 2023

Instructor: Dr. Nouman Azam

NOTES:

1. Attempt all questions.

1. Suppose the domain of the $Q(x,y,z)$ consists of the triplets x,y,z where $x = \{0, 1, 2\}$, $y = z = \{0, 1\}$. Write out these propositions using conjunction, disjunction and negation. (Points 9, CLO 1)

a. $\forall y Q(0,y,0) =$

b. $\exists x Q(x,1,1) =$

c. $\exists z \neg Q(0,0,z) =$

2. Consider the poset $(\{1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60\}, |)$, where $|$ is the divides relation. (Points 9, CLO 2)

a) Draw the Hasse diagram for the given poset.

b) Determine the minimum and maximum in the Hasse diagram.

c) Find the least and greatest in the Hasse diagram.

3. Circular Graphs (Points 9, CLO 3)

a) Draw a circular graph for the graph represented by set of vertices $= \{a,b,c,d\}$, set of edges $= \{(a,b), (c,d), (a,c), (b,d)\}$

b) A circular graph has 199 chords which are parallel to each other. How many vertices and edges will be there in the corresponding graph.

c) Draw a circular graph corresponding to the adjacency matrix below

	a	b	c	d
a	0	1	1	1
b	1	0	0	1
c	1	0	0	1
d	1	1	1	0

4. We need to schedule the final exam using the least number of time slots for the courses named as A, B, C, D, E. Moreover, we are given that there are no common students in the courses D & E. A & B, A & C, C & D. There are however common students in the remaining combination of the courses. Using Graph coloring, show the minimum number of time slots that are needed to conduct the exam. (Points 3, CLO 3).

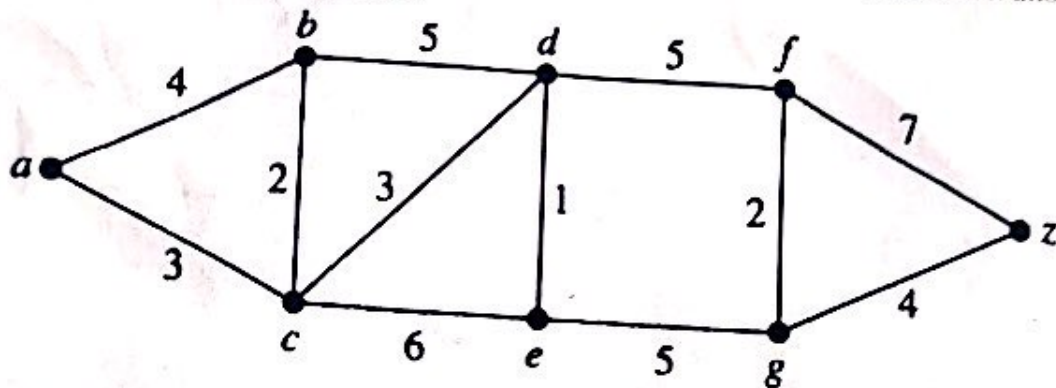
5. Construct the following (Points 6, CLO 3)

a) Adjacency matrix representing K_6

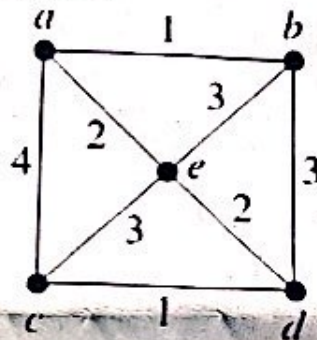
b) Incidence matrix representing $K_{1,3}$

6. An edge coloring of a graph is an assignment of colors to edges so that edges incident with a common vertex are assigned different colors. Use edge coloring to color the following graphs
(Points 6, CLO 3)
- K_6
 - C_6

7. Using Dijkstra algorithm, find the minimum path from node d to all other nodes. Show all the steps and the tables. (Points 5, CLO 3)



8. Consider the graph below (Points 6, CLO 4)



- Use Prim's algorithm to find a minimum spanning tree for the given weighted graphs. (The starting edge should be (c,d))
- Use Kruskal algorithm to find a minimum spanning tree for the given weighted graph. (The starting edge should be (c,d))

9. Compute the values of the following. (Points 6, CLO 4)

- Prefix Notation: $\cap \{a\} \cup \{a,b\} \cap \{a,b,c\} \{d,e,a,b\}$
- Postfix Notation: $\{2,3,4\} \{3,4,5\} \{4,3,5,7\} \{2,3,4\} \cap \cap \cap$

10. Use backtracking to solve the following problems 3-Queens (show the computation even if there is no solution) (Points 3, CLO 4)

Good Luck



Name: _____

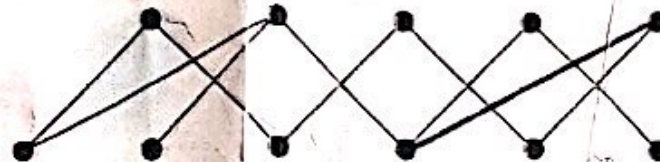
Program: BS (CS)
Semester: Fall - 2022
Time Allowed: 3: 00 hour
Course: Discrete Structures

Roll No: _____

Examination: Final
Weight 50% Total Marks: 53
Date: Dec, 2022
Instructor: Dr. Nouman Azam

NOTES:

1. Attempt all questions.



1. Graph basics (Points $2 \times 5 = 10$) [CLO 3]

- Determine the chromatic number of the graph shown above.
- What are the possible values for the chromatic number of a graph with 1000 vertices and one edge
- A circular graph has 199 chords which are parallel to each other. How many vertices and edges will be there in the corresponding graph.
- Draw a circular graph corresponding to the adjacency matrix below

	a	b	c	d
a	0	1	1	1
b	1	0	0	1
c	1	0	0	1
d	1	1	1	0

- Compute the degrees of the individual vertices in the graph above in part (a)

2. Construct the following (Points $2 \times 4 = 8$) [CLO 3]

- Adjacency matrix representing K_6
- Adjacency matrix representing Q_3
- Incidence matrix representing C_4
- Incidence matrix representing $K_{3,3}$

3. How many connected components are there in the graph with 1000 vertices (Points $2 \times 4 = 8$) [CLO 3]

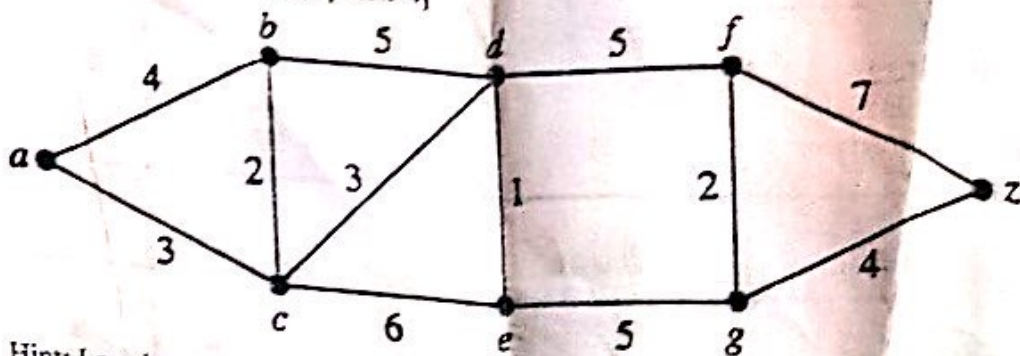
- Having one edge which is not a loop.
- Having two edges which are not loops and are between two distinct pair of vertices.
- Having 3 edges, assuming that the graph is a simple graph with no circuits. Moreover, the maximum length of any path in the graph is 1.
- Having 1 simple circuit of length 10 and no other edges. Moreover, assume that the graph is a simple graph

4. An edge coloring of a graph is an assignment of colors to edges so that edges incident with a common vertex are assigned different colors. Use edge coloring to color the following graphs (Points $2 \times 2 = 4$) [CLO 3]

a) K_6
b) C_6

✓ Note: I am talking about edges and not vertices. If you colored the vertices you receive zero.

5. Consider the weighted graph shown below. Use the Dijkstra algorithm to find the minimum path between f and a. (Points 5) [CLO 4]



Hint: I need:

1. A table which shows the application of the Dijkstra algorithm.
2. The minimum path
3. The minimum distance

- ✓ 6. Consider the graph in Question 5, construct the minimum spanning tree of the graph using (Points $2 \times 2 = 4$) [CLO 4]

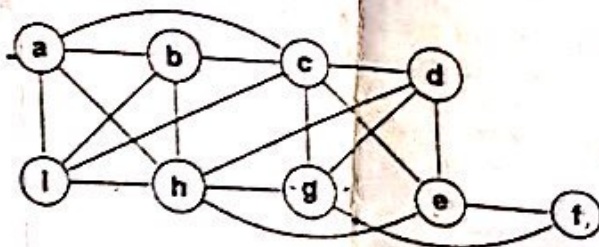
- a) Prim's algorithm (show me the final table)
- b) Using Kruskal's algorithm (show me the final table)

- ✓ 7. How many cut edge and cut vertices are there in the following graphs. First draw the graphs and then show the cut edges and cut vertices (Points $2 \times 2 = 4$) [CLO 4]

- a) C_4
- b) $K_{3,3}$

- ✓ 8. Use backtracking to solve the problems of (Points $2 \times 2 = 4$) [CLO 4]

- a) 5-Queens
- b) Find the subset, if it exists of the set $\{27, 24, 19, 14, 11, 8\}$ with a sum of 41.



- ✓ 9. Consider the graph above, determine if the graph has (Points $2 \times 2 = 4$) [CLO 4]
- a) Euler circuit, if yes, construct it and show it
 - b) Hamilton circuit, if yes, construct it and show it

- ✓ 10. Use backtracking to solve the following problems 3-Queens (show the computation even if there is no solution) (Points 2) [CLO 4]

Good Luck

Discrete Structure Final

Pa

Name:

Roll No:

Program: BS (CS, SE)

Semester: Fall – 2021

Time Allowed: 3 Hours

Course: Discrete Structures

Examination: Final

Weight 55% Total Marks: 110

Date: Jan, 2022

Instructor: Dr. Nouman Azam

NOTE:

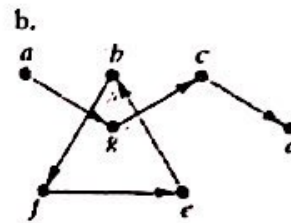
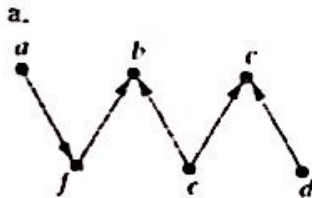
Attempt all questions on a sheet of paper.

Try attempting all questions in the order they are given in the paper.

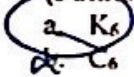
Graphs

1. Degree sequence: The degree sequence of a graph is the sequence of the degrees of the vertices of the graph in a decreasing order. For instance the degree sequence of W_5 is 5,3,3,3,3,3. What is the degree sequence of (Points $2 \times 8 = 16$)
 - a. What is the degree sequence of W_n
 - b. What is the degree sequence of $K_{m,n}$ ($m > n$)
 - c. What is the degree sequence of C_n
 - d. What is the degree sequence of K_n
 - e. Compute the edge in a graph with degree sequence 4, 3, 3, 2, 2
 - f. Compute the edge in a graph with degree sequence 5, 2, 2, 2, 2, 1
 - g. Draw the graph in part e
 - h. Draw the graph in part f
2. The complementary graph G of a simple graph G has the same vertices as G . Two vertices are adjacent in G if and only if they are not adjacent in G . (Points $2 \times 4 = 8$)
 - a. Draw the complementary graph corresponding to K_5 .
 - b. Draw the complementary graph corresponding to W_5 .
 - c. Construct the adjacency matrix for the graph in part a.
 - d. Construct the adjacency matrix for the graph in part b.
3. Circular graph (Points $2 \times 3 = 6$)
 - a. Draw circular graph for the graph represented by Set of vertices = $V = \{a_1, a_2, a_3, a_4\}$, Set of edges = $E = \{(a_1, a_2), (a_3, a_4), (a_1, a_3), (a_2, a_4)\}$
 - b. Draw circular graph for K_6 .
 - c. Draw circular graph for W_5
4. Construct the following (Points $2 \times 4 = 8$)
 - a. Adjacency matrix representing C_6
 - b. Adjacency matrix representing W_5
 - c. Incidence matrix representing C_4
 - d. Incidence matrix representing K_4
5. Answer the following questions (Points $2 \times 4 = 8$)
 - a. Does the graphs W_5 and C_5 has a euler path.
 - b. Does the graphs W_6 and C_6 has a euler circuit.
 - c. Does the graphs W_7 and C_7 has a hamilton path.
 - d. Does the graph W_8 and C_8 has a hamilton circuit

Determine whether each of these graphs are strongly connected and if not, whether it is weakly connected. (Points $2 \times 2 = 4$)



7. An edge coloring of a graph is an assignment of colors to edges so that edges incident with a common vertex are assigned different colors. Use edge coloring to color the following graph: (Points $4 \times 2 = 8$)



Partial Order

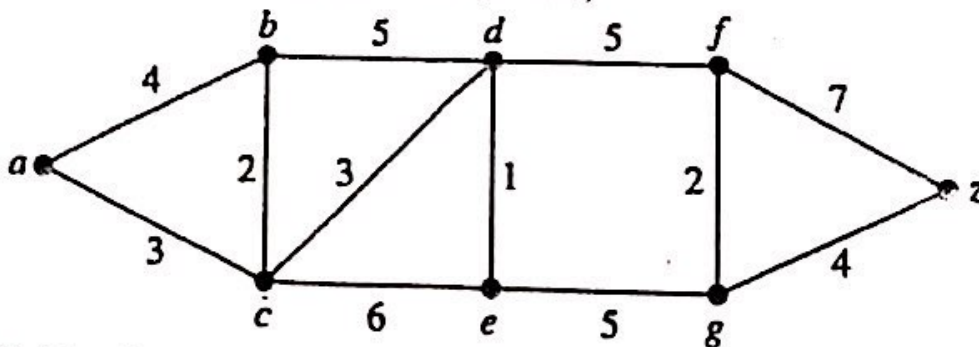
8. Consider the following partial order $((2, 3, 6, 12, 24, 36, 48, 60, 72), |)$ (Points $2 \times 5 = 10$)
- Define the partial order relation (Hint: I need a set containing tuples satisfying the relation)
 - Construct the corresponding Hasse diagram of the relation defined in part (a)
 - Is this a lattice. Explain
 - What is the greatest and least element if any exists.
 - What are maximal and minimal elements if any exist.
9. $(5+2+2+2+2+2+2+2 = 19)$
- Consider Hasse diagram with only four elements. Construct all possible types of Hasse diagrams that can be constructed. Note: I need only the diagrams.
 - Can we have a Euler circuit in a total order
 - Can we have a Euler path in the total order
 - Can we have simple circuit in a total order
 - A total order has only a single element. Do we have maximal element in such a total order?
 - Consider the total order of part e. Do we have a minimal element in such a total order?
 - Consider the total order of part e. Do we have a greatest element in such a total order?
 - Consider the total order of part e. Do we have a least element in such a total order?

Trees

10. Consider the following set theoretic expression $((Z \cup Y) - X) - (Z \cup (Y - Z))$ ($2 \times 4 =$ Points 8)
- $\cup$ = union and $-$ = set subtraction
- Represent the above expression using ordered rooted tree.
 - Write this expression in prefix notation.
 - Write this expression postfix notation.
 - Write this expression infix notation.

Shortest Paths and Spanning Trees

11. Consider the weighted graph shown below. Use the Dijkstra algorithm to find the minimum path between vertex 'g' to every other vertex. (Points 5)



Hint: I need

1. A table which shows the application of the Dijkstra algorithm.
2. The minimum path from 'g' to every other vertex
3. The minimum distance of 'g' to every other vertex

12. Consider the graph of W_6 and K_5 , construct the spanning-tree of these graphs using (Points $2 \times 2 = 4$)

- a. Depth first search algorithm.
- b. Breadth first search algorithm.

13. Use backtracking to solve the following problems (Points $2 \times 3 = 6$)

- ✶ 5-Queens
- ✶ Find the subset, if it exists of the set $\{27, 24, 19, 14, 11, 8\}$ with a sum of 41.

Good Luck

Name: _____

Roll No: _____

Program: BS (CS, SE)

Semester: Spring - 2020

Time Allowed: 3 Hours + 1 hour for uploading

Course: Discrete Structures

Examination: Final

Weight 60% Total Marks: 118

Date: June, 2020

Instructor: Dr. Nouman Azam

NOTE:

Attempt all questions on a sheet of paper.

Try attempting all questions in the order they are given in the paper.

Attempt on a paper using pen / pencil, scan and then send me in soft preferably through slate

And if slate is not working then through slack using direct message.

If both slate and slack are out of order then email me directly (consider this as a last option)

Make sure your paper is not visible to anyone else or it will be cheating

Relations

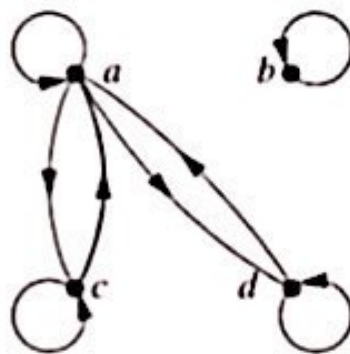
- Consider the relation R given by $\{(a,b) \mid 2a + 3b \text{ is divisible by } 5\}$, (Points $3 \times 3 = 9$)
 - Compute the relation R on defined on the set $A = \{1,2,3,4\}$
 - Determine if R is reflexive when it is define on the set $A = \mathbb{Z}^+$
 - Determine if R is reflexive when it is define on the set $A = \mathbb{Z}$
- Explain why the relation $R = \emptyset$ defined on a non empty set A , cannot be reflexive but is symmetric and transitive. (Points 3)
- Determine if the relation $R = \{(a,b) \mid a \text{ is taller than } b\}$ defined on the set of all people is (Points $3 \times 4 = 12$)
 - Reflexive
 - Symmetric,
 - Ant symmetric,
 - Transitive,

Note: Explain each part with proper definition and explanation
- Consider a relation R which is given based on the matrix given below. The relation is defined based on the set $A = \{1, 2, 3, 4\}$ and the rows and columns of the matrix corresponds to the integers in set A listed in increasing order. (Points $3 \times 3 = 9$)

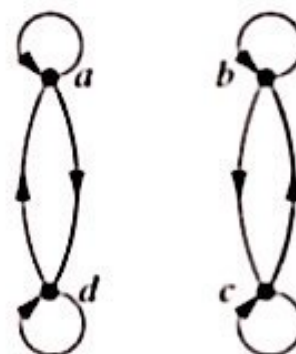
$$M_R = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$
 - Compute the tuples in R
 - Compute the tuples in R^{-1}
 - Compute the tuples in $R \cup R^{-1}$ and $R \cap R^{-1}$

5. Explain with reasons if the relations given below in the form of graphs are equivalence relations? If they are equivalence relations then provide the equivalence classes and partitions (Points 3*2 = 6)

a)



b)



6. Consider the following partial order $(\{2, 3, 6, 12, 24, 36, 48, 60, 72\}, |)$ (Points 5*3 = 15)
- Define the partial order relation (Hint: I need a set containing tuples satisfying the relation)
 - Construct the corresponding Hasse diagram of the relation defined in part (a)
 - Is this a lattice. Explain
 - What is the greatest and least element if any exists.
 - What are maximal and minimal elements if any exist.

Graphs

6. The degree sequence of a graph is the sequence of the degrees of the vertices of the graph in a decreasing order. What is the degree sequence of the following graphs (Points 3*4 = 12)
- W_5
 - $K_{3,3}$
 - C_6
 - K_6
7. Draw a circular graph for the graph represented by the following sets (Points 3)
- Set of vertices = $V = \{a_1, a_2, a_3, a_4\}$
 - Set of edges = $E = \{(a_1, a_2), (a_3, a_4), (a_1, a_3), (a_2, a_4)\}$
8. Draw a circular graph representing K_4 . (Points 3)
9. Draw the adjacency matrix for the graph $K_{2,3}$. Assume that the first group/set of vertices include $\{a, b\}$ and the second group/set of vertices include $\{c, d, e\}$ (Points 3)
Hint: One adjacency matrix is needed
10. Determine whether the graphs corresponding to the following two matrices are isomorphic or not (Points 3)

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

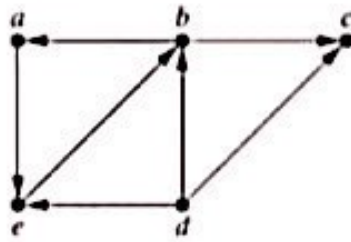
Hint: I need the 9 step procedure outlined in the class. You will first need to draw the graphs.

11. Answer the following questions with justification (Points $4 \times 3 = 12$)

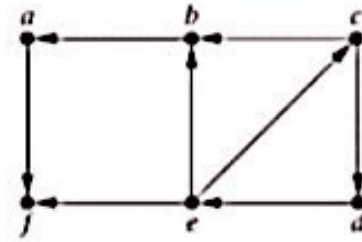
- Does the graphs W_N and C_N has a euler path when N is odd.
 - Does the graphs W_N and C_N has a euler circuit when N is even.
 - Does the graphs W_N and C_N has a hamilton path when N is even.
 - Does the graph W_N and C_N has a hamilton circuit when N is odd.
- Note: Each part must have two answers with justifications

12. Find the strongly connected components of each of the graphs given below (Points $3 \times 2 = 6$)

a.



b.



13. What is the chromatic number of a graph with (Points $3 \times 2 = 6$)

- K_N where N is odd.
- C_N where N is even.

Hint: Provide justifications for your answer

Trees

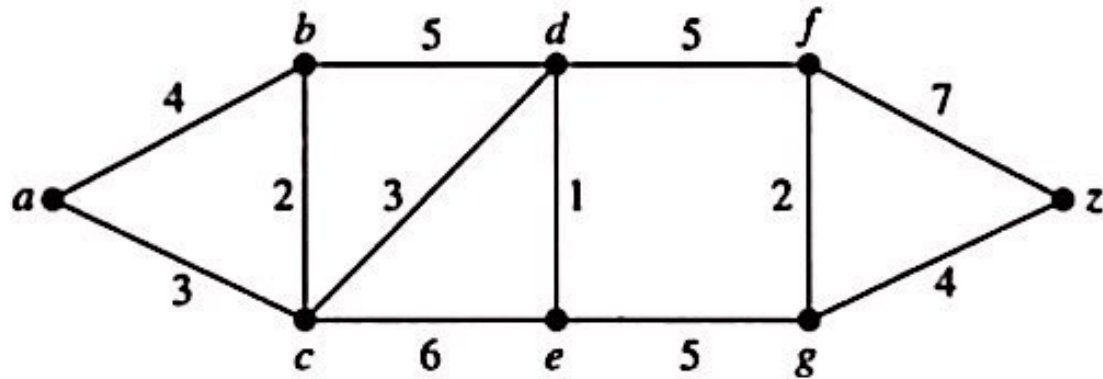
14. Consider the following set theoretic expression $((A + B) \cdot C) - (A \cdot (B - A))$ (Points $4 \times 3 = 12$)

- Represent the above expression using ordered rooted tree.
- Write this expression in prefix notation.
- Write this expression postfix notation.
- Write this expression infix notation.

Hint: It's a friendly question. In part (a) I expect a tree and in the (b – c) I expect expressions

Shortest Paths and Spanning Trees

15. Consider the weighted graph shown below. Use the Dijkstra algorithm to find the minim path between f and a. (Points 3)



Hint: I need

1. A table which shows the application of the dijistra algorithm.
 2. The minimum path
 3. The minimum distance
16. Consider the graph of W_5 and K_6 , construct the spanning tree of these graphs using (Points $3 \cdot 2 = 6$)
- g. Construct spanning trees of these graphs using depth first search.
 - h. Construct spanning trees of these graphs using breadth first search.

Good Luck

Name: M. Ahsan

Roll No: P17-6142

Program: BS (CS)

Examination: Final

Semester: Fall - 2018

Weightage 50% Total Marks: 59

Time Allowed: 3: 00 hour

Date: 2nd January, 2018

Course: Discrete Structures

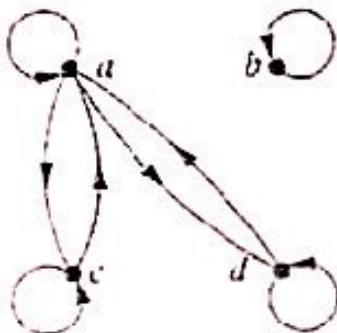
Instructor: Dr. Nouman Azam

NOTE: Attempt all questions.

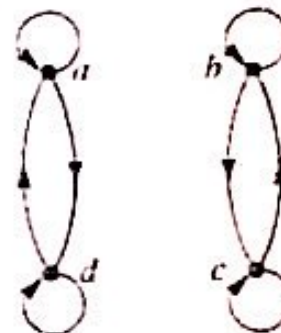
Equivalence Relation

1. Determine whether the relation represented by the following graphs are equivalence relations or not. Give the reasons (Points 2*2)

a)



b)



Hint: I need only the names of the properties which are satisfied/not satisfied based on which you are concluding that the relations are equivalence or not

2. Determine whether the relation represented by the following matrices equivalence relations or not. Give the reasons (Points 2*2)

a)

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

b)

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Hint: I need only the names of the properties which are satisfied/not satisfied based on which you are concluding that the relations are equivalence or not

Partial Order

3. Consider Hasse diagram with only three element. Construct all possible types of Hasse diagrams that can constructed. (Points 2)
Hint: There are less than 5 possibilities. Guess the number ☺. I need only the diagrams.
4. Consider the following partial order $(\{0, 2, 5, 10, 11, 15\}, \geq)$ (Points 5*1)
a) Define the partial order relation (Hint: I need a set containing tuples satisfying the relation)
b) Construct the corresponding Hasse diagram of the relation defined in part (a)
c) Is this a lattice (Hint: Yes / No)
d) What is the greatest and least element.
e) What is maximal and minimal element.

Common Graph Types and Bipartite Graph

5. How many vertices and how many edges does each of these graphs have (Points 5*2)
a) K_n b) C_n c) W_n d) $K_{m,n}$ e) Q_n
Hint: I need two numbers for each part
6. Are the following graphs bipartite (Points 1*2)
a) C_5 b) C_6

Representing Graphs and Graph Isomorphism

7. Draw the adjacency matrix for the graph $K_{3,3}$ (Points 2)
Hint: One adjacency matrix is needed
8. Determine whether the graphs corresponding to the following two matrices are isomorphic or not (Points 3)

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Hint: I need the 7 step procedure outlined in the class. You will first need to draw the graphs

Euler and Hamilton paths and circuits

9. For what minimum value of n , does the graph K_n has both Euler path and Euler circuit. (Points 2*1)
Hint: Tell me the value of n

10. Consider the graph below (Points 4*1)

- Does the graph has a euler path. If yes construct one.
- Does the graph has a euler circuit. If yes construct one.
- Does the graph has a hamilton path. If yes construct one.
- Does the graph has a hamilton circuit. If yes construct one.

Hint: The answers should contain two parts.

- Yes / No
- In case of yes, a path.



Connectedness and Graph Coloring

11. How many cut edges and cut vertices are there in the following graph (Points 4)



Hint: I need two numbers. This is your easy question. (haha☺)

12. What is the chromatic number of a graph with (Points 2*2)

- 5 vertices and no edge.
- 5 vertices and 1 edge.

Hint: I need one number in each case

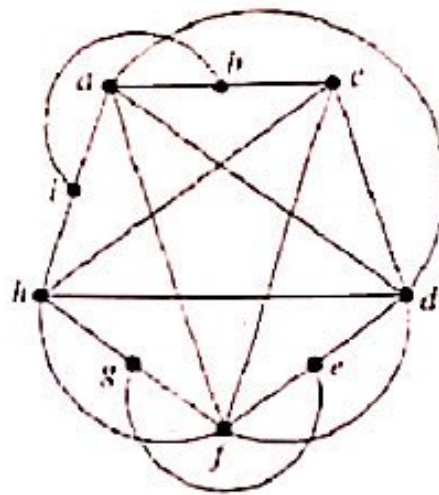
Planar Graphs

13. You are told that the following graph is planar since it is clear that it contains K_5 . Now using the theorem which states "A graph is planar if and only if it contains a subgraph homeomorphic to K_5 or K_3 ". Proof that the graph is homeomorphic to K_5 . (Points 2)

Hint: There are two steps.

Step 1: Obtain a subgraph from the original graph which you think is closer in shape to K_5

Step 2: Apply the elementary subdivision operation successively to obtain K_5



Tree Traversal

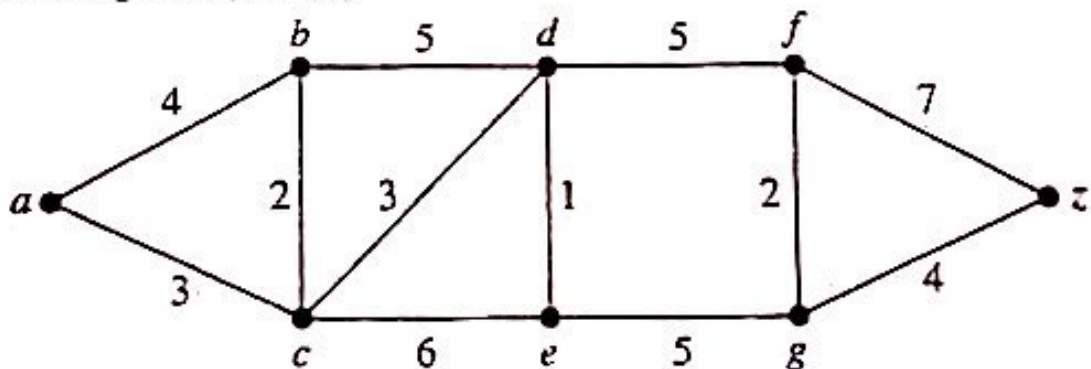
14. Consider the following set theoretic expression $(A \cap B) - (A \cup (B - A))$ (Points 4*1)

- Represent the above expression using ordered rooted tree.
- Write this expression in prefix notation.
- Write this expression postfix notation.
- Write this expression infix notation.

Hint: It's a friendly question. In part (a) I expect a tree and in the (b - c) I expect expressions

Shortest Paths and Spanning Trees

15. Consider the weighted graph shown below. Use the Dijkstra algorithm to find the minim path between g and b. (Points 3)



Hint: I need

- A table which shows the application of the dijkstra algorithm.
- The minimum path
- The minimum distance

16. Consider the graph of K_3 and C_6 , construct the spanning tree of these graphs using (Points 2*2)

- Construct spanning trees of these graphs using depth first search.
- Construct spanning trees of these graphs using breadth first search.

Good Luck